

A new screening method for personal identification using dental optical impression data

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Abstract

Most personal identification techniques using dental forensics involve a direct comparison of analog images of the oral cavity with preexisting dental charts. Nevertheless, for the forensic identification of individuals, more efficient and faster methods are required. The progression in digital technology in dentistry has been remarkable, and oral scanning and digital X-ray imaging devices are now used routinely in dental practice. We report a new personal identification technique using intraoral Standard Triangled Language (STL) data, which were obtained using a digital intraoral scanner. We compared the STL data from two virtual corpses with those from a larger population of dental clinic patients to investigate whether the corpses could be identified based solely on the dental impression data. We applied the following two methods: analysis of the area of the three virtual planes connecting the cusp tip of the three adjacent molars (method A) and analysis of the area of the virtual three planes connecting the bulge of each of the three molars (method B). No virtual corpse could be identified with either of the methods with a margin of error of 0 mm². Method A was able to perform the screening without omitting the individual within a margin error of $\geq \pm 4$ mm² to obtain a positive result from the screening. In method B, the error range had to be increased to $\geq \pm 5$ mm². Method A was able to search more accurately than method B, possibly because of the large standard deviation of the measured values across the two teeth. Method B might require multiple measurements to obtain accurate search results. In conclusion, we suggest that our new method could improve the efficiency of personal identification.

Key words: disaster victim identification, dental findings, forensic dentistry, personal identification, surface triangled language data

Introduction

The personal identification of corpses can be performed on the basis of their belongings and physical characteristics. Fingerprints, palm prints, DNA typing, and dental findings are reliable and scientifically accurate methods used for identification. One-on-one identification, such as identification through a parent–child relationship, is an option that has a high degree of certainty¹. However, in the event of a disaster, it takes time and effort to search for the DNA typing of a family member among a large number of people. Moreover, the DNA typing of an individual or their family member is known in only a few cases, and the centralized management of this information is a challenge. In Japan, collecting DNA data is problematic because of

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issues with personal information protection (i.e., DNA data may provide information on illnesses or genetic mutations). Additionally, in the event of a disaster, stored information may be lost due to destruction or deterioration².

Because the teeth and jawbone are hard tissues, they survive even under harsh conditions. Hence, in the event of a mass disaster, the use of dental information is an effective method for comprehensive personal identification.

Dental findings can be collected directly from bodies. Thereafter, personal identification is possible by collating the dental findings using dental treatment records, dental X-rays, and gypsum models³⁻⁵. This information is generally stored in dental clinics^{2,3}. After the 1985 Japan Airlines crash, dental personal identification received a great deal of attention, and it has been used for definitive personal identification for the past 35 years in Japan.

In the 2005 Indian Ocean earthquake and tsunami disaster, the identification rate of missing persons with dental records was significantly higher than that of people without dental records^{6,7}. As of March 2021, the number of deaths from the Great East Japan Earthquake was 13,373, of which approximately 8% were recognized because of dental findings. In some cases, personal identification was possible because of the type of oral implant used, the site of implantation, and surgical history⁸.

By contrast, in the 2012 Lagos-Dana aircraft crash in Nigeria, 97.4% of individuals were identified by a combination of DNA typing and dental findings. However, as there were no dental findings before the death of the unidentified persons, only <10% of them were identified by forensic dentistry⁹.

Personal identification based on dental findings is useful; however, to date, this method has been based primarily on the analog collation of dental charts. Thus, it is dependent on the limited number of personnel available at the time of a major disaster. There is a need to improve the work efficiency and enhance the safety and certainty of the process. Additionally, in the Great East Japan Earthquake, it was extremely difficult to verify the identity of individuals because of the loss of dental treatment records and dental X-rays resulting from the tsunami. In conclusion, data storage, rather than data collection, has been an issue¹⁰.

Thus, antemortem digital data of dental findings could be a better alternative. Dental X-ray images of the teeth contain specific individual information, similar to DNA typing, and are considered to be

highly useful for personal identification¹¹. However, to convert the information from dental X-ray images into data, dentists must perform registration work. From the standpoint of personal information, there are major barriers to personal information management and storage. It is possible that these issues can be solved by screening the antemortem digital dentition data of corpses. Thus, this study aimed to identify a more efficient method of performing personal identification.

Materials and methods

Target sample and virtual corpse settings

In this study, we included randomly collected intraoral gypsum models of patients who visited two dental clinics between December 1, 2020, and August 1, 2021. Additionally, two intraoral gypsum models of the authors of this paper were collected as virtual corpses X and Y. All gypsum models were set as population data. This study was approved by the Showa University Medical Ethics Committee (approval No. 2020-005).

Digitalization of the gypsum models

We obtained optical impressions from all gypsum models using intraoral scanners (DORAOCs-IC25, DIGITAL PROCESS LTD, Japan). The obtained Standard Triangled Language (STL) data were processed using image processing software (exocad, DentalDB for DIPRO, version 2015.07, Modern UI, Forest-One Co., Ltd., Japan). These data were saved as population data in STL format.

Digitalization of oral dentition of the virtual corpses

We obtained intraoral optical impressions from virtual corpses X and Y using an intraoral scanner (Trophy 3DI Pro, Trophy radiology Japan CO., LTD.). These data were saved in STL format.

Parameter settings for personal identification

The parameter for identification of corpses X and Y was set as the area defined by the line that connects the cusp tip and the bulge of each tooth. In this study, we used the following two methods to compare and examine the personal identification methods devised. In method A, the area of the virtual plane created by connecting the cusp tips of the adjacent teeth was chosen as the identification parameter. Specifically, area “a” was defined as the area of the virtual plane formed by the buccal cusp of the first premolar, the buccal cusp of the second

premolar, the lingual cusp of the second premolar, and the lingual cusp of the first premolar. The area of the virtual plane formed by the buccal cusp of the second premolar, the mesial buccal cusp of the first molar, the mesial lingual cusp of the first molar, and the lingual cusp of the second premolar was defined as “b.” Area “c” was defined as the area of the virtual plane formed by the mesial buccal cusp of the first molar, the distal buccal cusp of the first molar, the distal lingual cusp of the first molar, and the mesial lingual cusp of the first molar was (Figure 1).

The area of the virtual plane created from the bulge of a single tooth was used as the identification parameter in method B. Specifically, the area of the virtual plane formed by the buccal bulge of the first premolar, the distal bulge of the first premolar, the lingual bulge of the first premolar, and the mesial bulge of the first premolar was defined as “d.” The area of the virtual plane formed by the buccal bulge of the second premolar, the distal bulge of the second premolar, the lingual bulge of the second premolar, and the mesial bulge of the second premolar was defined as “e.”

The area of the virtual plane formed by the buccal bulge of the first molar, the distal bulge of the first molar, the lingual bulge of the first molar, and the mesial bulge of the first molar was defined as “f” (Figure 2). The area was calculated as follows: $1/2 \times \alpha \times \beta \times \sin \theta$, where α and β are the distances connecting the diagonal lines of the squares and θ is the angle at which α and β intersect. We used GOM Inspect Suite (version 2.0.1, ZEISS Company, Germany) to set and measure each parameter (Figure 3).

We stored data from each area for each patient individually. When the area could not be obtained because of the loss of teeth, no data were recorded.

Search for virtual corpses and evaluation of parameter settings

We searched the population data using the parameters defined above for the virtual corpses X and Y. In methods A and B, the search was performed under the following conditions: a only, b only, c only, a and b, a and c, b and c, and a, b, and c. Furthermore, because technical differences can occur when acquiring optical impression data and in the parameter settings, a margin of error was provided for the parameters of the virtual corpses X and Y. We set the margin of error to ± 1 , ± 2 , ± 3 , ± 4 , ± 5 , and $\pm 6 \text{ mm}^2$.

We searched and summarized the data using Microsoft Excel 2019. The numbers in the graph

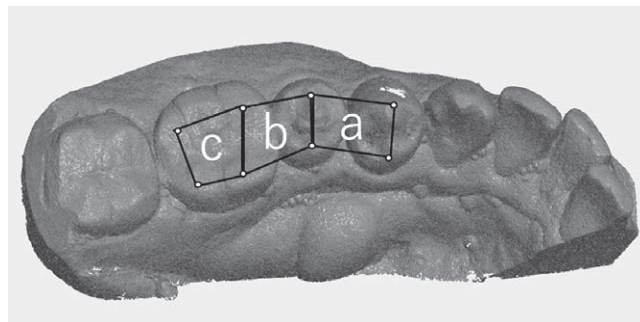


Fig. 1. In method A, the area of the virtual plane created by connecting the cusp tips of adjacent teeth was chosen as the identification parameter.

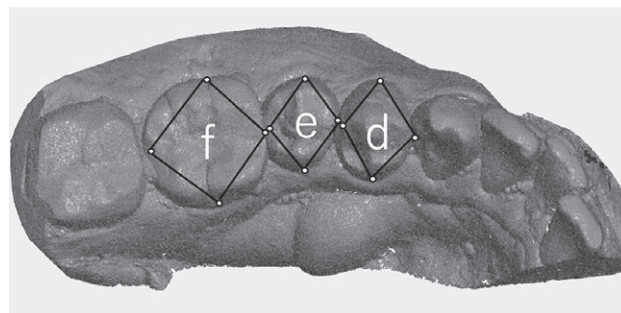
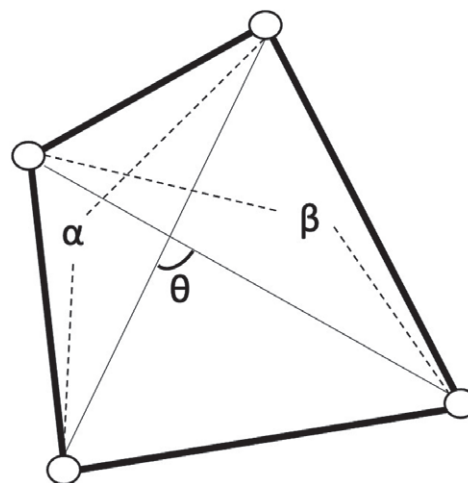


Fig. 2. In method B, the area of the virtual plane created from the bulge of a single tooth was used as the identification parameter.



$$S = 1/2 \times \alpha \times \beta \times \sin \theta$$

Fig. 3. The area was calculated as follows: $1/2 \times \alpha \times \beta \times \sin \theta$, where α and β are the distances connecting the diagonal lines of the squares and θ is the angle at which α and β intersect.

represent the discrepancy rate (%), and the numbers in parentheses represent the number of searches. The larger the discrepancy rate, the higher the screening accuracy.

The numbers in the graph represent the discrepancy rate (%), and the numbers below the nonconformity ratio (%) represent the number of detections, including virtual corpse X or Y. The larger the discrepancy rate, the higher the screening accuracy. Results that did not include virtual corpse X or Y had a discrepancy rate of 0%. We performed the same procedure for method B. Consequently, we examined the possibility of a screening method in which virtual corpses X and Y agree.

Results

Target age and sex

We included optical dental impression data from 54 men and 49 women in this study. The overall average age of the participants was 46 years (men, 43 [22–84] years; women, 49 [20–96] years).

Search for virtual corpses (method A)

Both virtual corpses X and Y were $\pm 0 \text{ mm}^2$ under all conditions, and the subjects were not identified (Figure 4). The number of positive results increased as the error margin increased. Virtual corpse X could be screened only after introducing an error of $\pm 4 \text{ mm}^2$ or greater. Similar results were obtained for virtual corpse Y. We found no differences in sex or age (Table 1).

Search for virtual corpses (method B)

For virtual corpses X and Y, no individuals were detected without the introduction of the error margin under all conditions (Figure 5). The number of detections increased with the error. Virtual corpse X could be screened with certainty only with an error of $\pm 5 \text{ mm}^2$ or greater. Virtual corpse Y could be screened only with an error of $\pm 4 \text{ mm}^2$ or greater. The condition in which both corpses were identified was with an error margin of $\pm 5 \text{ mm}^2$ or greater. We found no differences in sex or age (Table 2).

Discussion

In this study, we performed a search of unidentified corpses without regard to sex or status of the upper and lower jaws. Advances in digital technology have been applied in dentistry. Digital devices, such as intraoral scanners and X-ray imaging devices, are currently being used in dental clinics. Additionally,

there has been a rapid increase in the number of cases in which the patient's oral data are stored as digital information. Particularly, the development of computer-aided design and manufacturing has promoted the widespread implementation of optical impression acquisition by an intraoral scanner. Hence, prosthetic devices can be fabricated without the need to create dental gypsum models or reproduce the patient's oral cavity using digital data^{12,13}. Various methods exist for personal identification, including fingerprinting and DNA typing, but the availability of these data is limited antemortem. DNA typing is particularly useful for personal identification, but it is difficult to collect and manage DNA that contains genetic information without obtaining the individual's consent. By contrast, according to a report by the Ministry of Health, Labor, and Welfare, the rate of dental consultations in Japan has been increasing since 2016 at a rate of 52.9%, and >60% of Japanese people have a designated dentist.

The results of this study suggest that STL data can be used for personal identification. STL data stored in dental clinics and laboratories are valuable information suitable for personal identification. Considering the rate of dental consultations in Japan, the STL data used in dentistry are likely to be the largest backup data for personal information. Digitalization in dentistry is progressing rapidly, and the information acquired from intraoral scanners is saved as digital data each time. In the event of an emergency, these data will become important dental information and social property. Although the use of optical impressions in Japan is lower than that in other countries, the rate is expected to increase in the future. Thus, the method used in this study is likely to be effective. Presently, a precise optical impression is limited to each tooth level. However, technological development will enable a precise impression of the entire oral cavity in a short time, and its use will likely spread.

The method used in this study does not consist of personal identification by 3D–3D collation but is rather based on measuring specific areas in the dental cavity and comparing the data with a population database. This study suggested the feasibility and high utility of this method. Additionally, the data obtained are suitable for storage because the amount of numerical data is smaller than that of STL. The backup of STL data is stored not only in dental clinics but also in dental laboratories. Even if the clinic is damaged by an earthquake, for instance, the data are stored in the cloud, and there is no risk of data loss.

In comparison with method B, results from

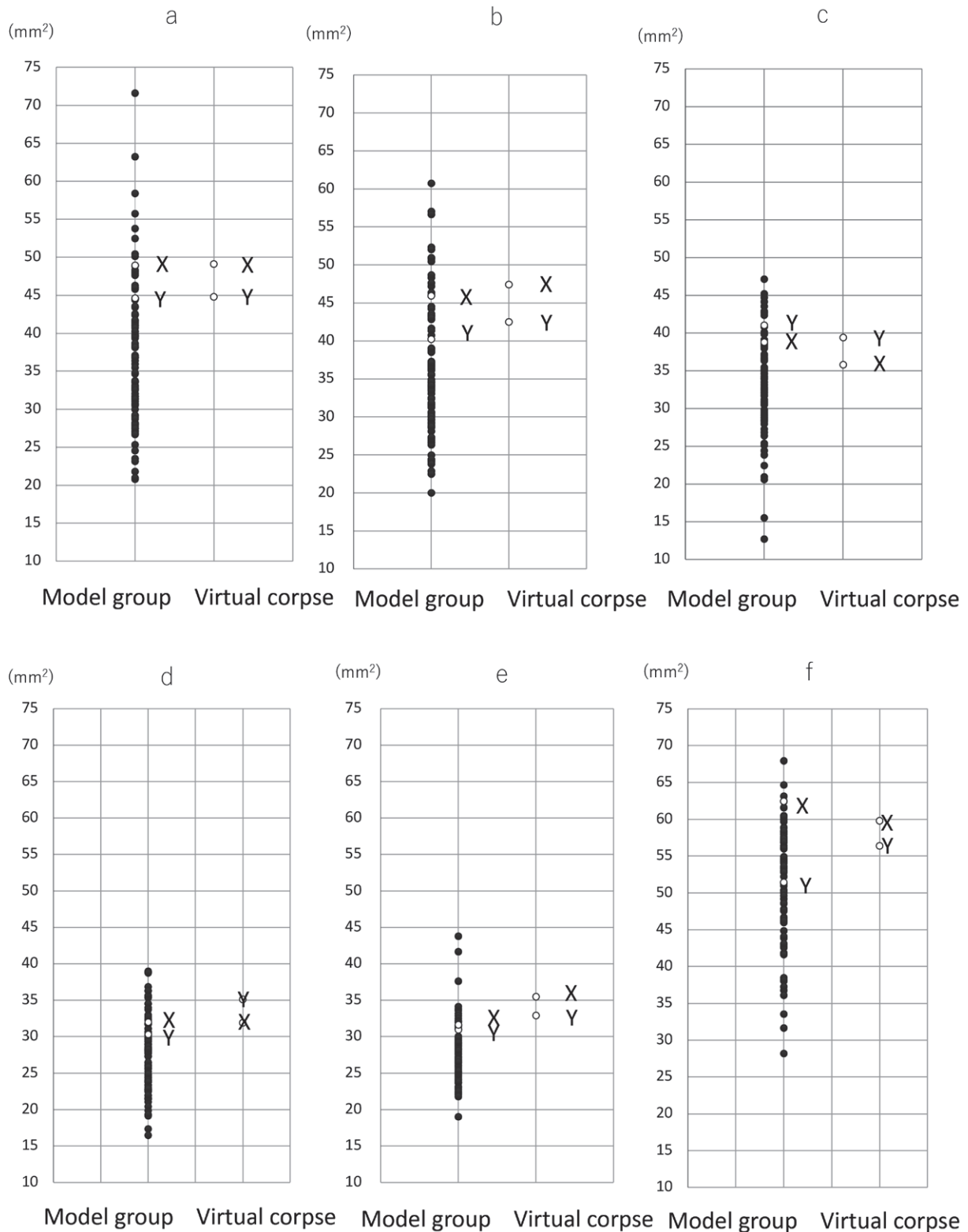


Fig. 4. 5. Both methods A and B had errors in the model group and virtual corpse. Therefore, screening was performed including errors. The vertical axis on the left represents the model group, and the vertical axis on the right represents the virtual corpse, indicating that the X and Y numbers in the model group and the X and Y numbers in the virtual corpse do not match exactly.

Table 1. The condition for not denying X and Y under all conditions was an error of $\pm 4 \text{ mm}^2$ or more

	Search item \ Error	$\pm 0 \text{ mm}^2$	$\pm 1 \text{ mm}^2$	$\pm 2 \text{ mm}^2$	$\pm 3 \text{ mm}^2$	$\pm 4 \text{ mm}^2$	$\pm 5 \text{ mm}^2$	$\pm 6 \text{ mm}^2$
Virtual Corpse X	a only	0%	96% 4	92% 8	90% 10	87% 13	82% 19	80% 21
	b only	0%	96% 4	93% 7	90% 9	83% 18	82% 19	78% 23
	c only	0%	0%	0%	0%	56% 45	46% 55	38% 64
	b and c	0%	0%	0%	0%	89% 11	85% 15	83% 18
	a and b and c	0%	0%	0%	0%	97% 3	95% 5	91% 9
Virtual Corpse Y	a only	0%	93% 7	88% 12	83% 18	77% 24	67% 34	62% 39
	b only	0%	0%	90% 10	87% 13	83% 17	79% 22	70% 31
	c only	0%	0%	0%	0%	67% 34	57% 44	50% 52
	b and c	0%	0%	0%	0%	93% 7	88% 12	80% 21
	a and b and c	0%	0%	0%	0%	95% 5	91% 9	89% 11

Table 2. The condition for not denying X and Y under all conditions was an error of $\pm 5 \text{ mm}^2$ or more

	Search item \ Error	$\pm 0 \text{ mm}^2$	$\pm 1 \text{ mm}^2$	$\pm 2 \text{ mm}^2$	$\pm 3 \text{ mm}^2$	$\pm 4 \text{ mm}^2$	$\pm 5 \text{ mm}^2$	$\pm 6 \text{ mm}^2$
Virtual Corpse X	d only	0%	0%	77% 24	64% 37	56% 45	50% 52	42% 60
	e only	0%	0%	0%	0%	78% 23	76% 25	67% 34
	f only	0%	0%	0%	0%	73% 28	63% 38	56% 28
	d and e	0%	0%	0%	0%	89% 11	84% 16	73% 28
	e and f	0%	0%	0%	0%	0%	87% 13	82% 19
	d and e and f	0%	0%	0%	0%	0%	90% 10	84% 16
Virtual Corpse Y	d only	0%	0%	0%	0%	76% 25	70% 31	63% 38
	e only	0%	84% 16	80% 21	70% 31	59% 42	53% 48	42% 60
	f only	0%	0%	0%	0%	62% 39	58% 43	51% 50
	d and e	0%	0%	0%	0%	82% 19	76% 25	72% 29
	e and f	0%	0%	0%	0%	81% 20	76% 25	67% 34
	d and e and f	0%	0%	0%	0%	94% 6	89% 11	84% 16

Table 3. Compared with method B, results from method A have a large standard deviation because the measurement spans two teeth; thus, a more accurate screening was possible

	Method A	Method B
Discrepancy rate	High	Low
Tooth loss effect	Low	High
Attrition effect	Low	High
Orthodontics effect	Low	High

method A have a large standard deviation, because the measurement spans two teeth, enabling a more accurate screening (Table 3).

Our study suggests that both methods require a search that includes a margin of error. In method B, because each tooth was measured, the standard deviation was small, and data on multiple teeth were required for accurate screening. Regarding tooth morphology, it has been reported that in Sweden, determining the sex of 19-year-olds can be based on an analysis of tooth number measurement of the maxillary canines¹⁴. Similarly, a study conducted on the Portuguese population reported the means of sexual determination by analyzing the maxillary canines¹⁵. Moreover, there is a sex difference in the morphology of the molar teeth crown in males and females. In this study, we calculated the area of the cusp tip and the area of the bulge of the first premolar, second premolar, and first molar, but no differences in sex or age were observed for these parameters. Further analysis is required to determine whether this result is specific to Japanese subjects or only to this research target group.

Before now, personal identification using dental findings had to be examined case by case by a dentist. This is because only a dentist could read the findings in the oral cavity. Additionally, the collection and analysis of dental information for personal identification at scale is a laborious and time-intensive process. This study suggests that anyone, even those who are not dentists, can easily perform personal identification. However, to convert the information of the dental X-ray images into data, registration work by a dentist is required. Additionally, at this stage, there are few reports of the use of dentition/intraoral morphology for personal identification. Reportedly, in the event of a larger natural disaster in the future, digitization of antemortem data is important for personal identification of victims, thereby improving

the efficiency of data collation¹⁶. We believe that STL data-converted intraoral information will be used for dental personal identification. Moreover, by using cloud technology to register optical impression data electronically, it is possible to create a tool in which necessary information can be retrieved at any time. In preparation for a major disaster that could occur in the future, we will conduct further research to preserve optical impression information for a long period of time and use it to identify unknown corpses.

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Conflict of Interest disclosure

The authors declare no conflicts of interest associated with this manuscript.

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